

In this paper, we review in typical spaces in the general topology, from the language of functions.

Def.

Given a set X and a family of Topological space $\{Y_i\}_{i \in I}$ with functions $\{f_i: X \rightarrow Y_i\}$,

the topology $\mathcal{L}$ on X is called the 'initial topology' if

$(X, \mathcal{L})$ is the coarsest Topology such that for all $i \in I$, f_i is continuous

Equivalently,

$$\mathcal{L} = \bigcap \{ \mathcal{L}' \mid \text{for all } i \in I, f_i: (X, \mathcal{L}') \rightarrow Y_i \text{ is continuous} \},$$

and generated by the subbasis $\mathcal{S} := \{f_i^{-1}[U] \mid \text{for some } i \in I, U \subseteq Y_i \text{ open}\}$.

Dually, a set X and a family of Topological space $\{Y_i\}_{i \in I}$ with functions

$$\{g_i: Y_i \rightarrow X\}$$

the topology $\mathcal{L}$ on X is called the final topology if

$(X, \mathcal{L})$ is the finest Topology such that for all $i \in I$, g_i is continuous

Equivalently, the $\mathcal{L}$ is final topology if and only if $\mathcal{L}$ satisfies

$U \subseteq X$ is open set if and only if for each $i \in I$, $g_i^{-1}[U] \subseteq Y_i$ is open.

Moreover, given topological space Z and a map $h: Z \rightarrow X$, and $w: X \rightarrow Z$,

$h: Z \rightarrow (X, \mathcal{L}_{\text{init}})$ is continuous if and only if $f_i \circ h: Z \rightarrow Y_i$ is continuous, for all $i \in I$

$w: (X, \mathcal{L}_{\text{final}}) \rightarrow Z$ is continuous if and only if $w \circ g_i: Y_i \rightarrow Z$ is continuous for all $i \in I$.

Diagrammatically,

$$\begin{array}{ccc} Z & \xrightleftharpoons[w]{h} & X \xrightleftharpoons[g_i]{f_i} Y_i \end{array}$$

$\overline{\text{Product Space}}$

$T_{\text{sub space}}$

Metric space

Let X be a set with a Metric $d: X \times X \rightarrow \mathbb{R}$.

The Topology $\mathcal{E}_d$ generated by a basis $\{B_r(x) \mid x \in X, r > 0\}$

is the Coarser Topology such that the metric d is Continuous.

Proof. Given a basis element $(a, b) \subset \mathbb{R}$, $d^{-1}[(a, b)] = \{(x, y) \in X \times X \mid d(x, y) \in (a, b)\}$.

If $b \leq 0$, then the Preimage by d is empty, so open. Now, assume that $b > 0$.

Let $(x, y) \in d^{-1}[(a, b)]$ be given. Then, $a < d(x, y) < b$.

take $r_x, r_y > 0$ such that $r_x + r_y < \min\{d(x, y) - a, b - d(x, y)\}$.

(It is possible, rigorously, take $r_x = r_y = \frac{1}{2} \cdot \min\{d(x, y) - a, b - d(x, y)\}$.)

Then, given $(x', y') \in B_{r_x}(x) \times B_{r_y}(y)$, that is, $d(x, x') < r_x$ and $d(y, y') < r_y$,

By the triangle inequality gives that

$$d(x', y') \leq d(x', x) + d(x, y) + d(y, y') < r_x + d(x, y) + r_y < b$$

and

$$d(x, y) \leq d(x, x') + d(x', y') + d(y', y) < r_x + d(x', y') + r_y$$

$\Rightarrow a < d(x, y) - (r_x + r_y) < d(x', y')$, so that

$$a < d(x', y') < b, \text{ it means } (x', y') \in d^{-1}[(a, b)].$$

(We also used the symmetry of metric), therefore

$$(x, y) \in B_{r_x}(x) \times B_{r_y}(y) \subseteq d^{-1}[(a, b)].$$

Since $B_{r_x}(x) \times B_{r_y}(y)$ is open set of $X \times X$, so $d^{-1}[(a, b)]$ is open. $\therefore d$ is Continuous.

To show that the second assertion, let $\mathcal{E}'$ be a Topology on X such that the Metric $d: X \times X \rightarrow \mathbb{R}$ is continuous. Let $x_0 \in X$ be fixed.

Since every Continuous map is Continuous in each Variable separately, the map $f_{x_0}: X \rightarrow \mathbb{R}; x \mapsto d(x, x_0)$ is Continuous. We want to show that: $\mathcal{E}_d \subseteq \mathcal{E}'$.

Let $B_r(x) \in \mathcal{E}_d$ be an open set of the metric space. Then, for f_x ,

$$B_r(x) = f_{x_0}^{-1}[(-\infty, r)]$$

is open set of $\mathcal{E}'$, since f_x is continuous on $\mathcal{E}'$ and $(-\infty, r) \subseteq \mathbb{R}$ is open set. $\square$

the initial topology with respect to $\{d_x: X \rightarrow \mathbb{R} : Y \mapsto d(x, Y) \mid x \in X\}$.